Examining Active Region Features of the Lower Solar Atmosphere

In this tutorial, we will examine images of a solar active region at different wavelength passbands:

- optical continuum (SDO/HMI)
- 1600 \AA continuum (SDO/AIA)
- He II 304 \AA (He II Lyman alpha line; SOD/AIA)

and consider what we can infer about the different features present in each based on our understanding of the solar atmosphere.

Install relevant packages

In [ ]:
```
!pip install -q astropy
!pip install -q sunpy[all]
!pip install -q matplotlib
```

Import relevant packages

In [ ]:
```
import glob
import os.path

import astropy.units as u
import drms
import matplotlib.pyplot as plt
import sunpy.map
import sunpy.visualization.colormaps.cm as cm

from sunpy.net import Fido, attrs as a
from sunpy.time import TimeRange
```

Acquire Data

Acquire SHARP files (SHARP stands for Spaceweather HMI Active Region Patch)

In [ ]:
```
series = 'hmi.sharp_cea_720s'
sharpnum = 377 #sharp number
segments = ['magnetogram', 'continuum']
kwlist = ['T_REC', 'LON_FWT', 'OBS_VR', 'CROTA2',
          'CRPIX1', 'CRPIX2', 'CDELT1', 'CDELT2', 'CRVAL1', 'CRVAL2']

c = drms.Client(email="daniel.ryan@ucl.ac.uk") ##Use your own email addre
k = c.query('%s[%d]' % (series, sharpnum), key=kwlist, rec_index=True)

rec_cm = k.LON_FWT.abs().idxmin()
```

```python
k_cm = k.loc[rec_cm]
t_cm = drms.to_datetime(k.T_REC[rec_cm])
print(rec_cm, '@', k.LON_FWT[rec_cm], 'deg')
print('Timestamp:', t_cm)

t_cm_str = t_cm.strftime('%Y%m%d_%H%M%S_TAI')
fname_mask = '{series}.{sharpnum}.{tstr}.{segment}.fits'
fnames = {
    s: fname_mask.format(
        series=series, sharpnum=sharpnum, tstr=t_cm_str, segment=s)
    for s in segments}

download_segments = []
for k, v in fnames.items():
    if not os.path.exists(v):
        download_segments.append(k)

if download_segments:
    exp_query = '%s{%s}' % (rec_cm, ','.join(download_segments))
    r = c.export(exp_query, method='url', protocol='fits')
    dl = r.download('.')
```

Read SHARP data into sunpy maps.

In [ ]:
```python
# Define file name.
file_mag = glob.glob("*magnetogram.fits")

# Read file/data into a sunpy Map object.
map_mag = sunpy.map.Map(file_mag)
```

In [ ]:
```python
# Let's look at a summary of the sunpy Map
map_mag
```

In [ ]:
```python
# Define file name.
file_cont = glob.glob("*continuum.fits")

# Read file/data into a sunpy Map object.
map_cont = sunpy.map.Map(file_cont)
```

In [ ]:
```python
# Let's look at a summary of the sunpy Map
map_cont
```

Acquire corresponding AIA images

Now let's use the Fido subpackage of sunpy to find and download AIA images of the same active region.

In [ ]:
```python
# Define a time range to search based on the time of HMI images.
# Let's give a time range of +/- 5 seconds around the HMI time, using ast
search_timerange = TimeRange(map_mag.date-5*u.s, map_mag.date + 5*u.s)
```

1600 A

In [ ]:
```
# Find the AIA 1600 image corresponding to the same time as the above HMI
query_1600 = Fido.search(a.Time(search_timerange), a.Instrument.aia,
                         a.Wavelength(1600*u.AA))
```

In [ ]:
```
# Now let's use Fido to download the file to the current directory.
file_1600 = Fido.fetch(query_1600, path="./")
```

In [ ]:
```
# Read AIA 1600 file and crop it to the same FOV as the HMI images.
map_1600 = sunpy.map.Map(file_1600)
map_1600 = map_1600.submap(map_mag.bottom_left_coord,
                           top_right=map_mag.top_right_coord)
```

In [ ]:
```
# Let's get a summary of the contents of the 1600 Map.
map_1600
```

304 A

In [ ]:
```
# Find the AIA 304 image corresponding to the same time as the above HMI
query_304 = Fido.search(a.Time(search_timerange), a.Instrument.aia,
                        a.Wavelength(304*u.AA))
```

In [ ]:
```
# Now let's use Fido to download the file to the current directory.
file_304 = Fido.fetch(query_304, path="./")
```

In [ ]:
```
# Read AIA 304 file and crop it to same FOV as HMI maps.
map_304 = sunpy.map.Map(file_304)
map_304 = map_304.submap(map_mag.bottom_left_coord,
                         top_right=map_mag.top_right_coord)
```

In [ ]:
```
# Let's get a summary of the contents of the 304 Map.
map_304
```

Compare intensity features at different wavelengths with magnetic features

Calculate Contours of Magnetogram with 150G cut-off

In [ ]:
```
lvl = 150
pos_contours = map_mag.find_contours(lvl)
neg_contours = map_mag.find_contours(-lvl)
```

Plot Images

In [ ]:
```
linewidth = 0.65

fig = plt.figure(figsize=(12, 18))

ax_mag0 = fig.add_subplot(421, projection=map_mag)
map_mag.plot(axes=ax_mag0, vmin=-1000, vmax=1000)

ax_mag  = fig.add_subplot(422, projection=map_mag)
map_mag.plot(axes=ax_mag)
for contour in pos_contours:
```

```
    ax_mag.plot_coord(contour, color="blue", linewidth=linewidth)
for contour in neg_contours:
    ax_mag.plot_coord(contour, color="red", linewidth=linewidth)

ax_cont0 = fig.add_subplot(423, projection=map_cont)
map_cont.plot(axes=ax_cont0)

ax_cont = fig.add_subplot(424, projection=map_cont)
map_cont.plot(axes=ax_cont)
for contour in pos_contours:
    ax_cont.plot_coord(contour, color="blue", linewidth=linewidth)
for contour in neg_contours:
    ax_cont.plot_coord(contour, color="red", linewidth=linewidth)

ax_1600_0 = fig.add_subplot(425, projection=map_1600)
map_1600.plot(axes=ax_1600_0)

ax_1600 = fig.add_subplot(426, projection=map_1600)
map_1600.plot(axes=ax_1600)
for contour in pos_contours:
    ax_1600.plot_coord(contour, color="blue", linewidth=linewidth)
for contour in neg_contours:
    ax_1600.plot_coord(contour, color="red", linewidth=linewidth)

ax_304_0 = fig.add_subplot(427, projection=map_304)
map_304.plot(axes=ax_304_0)

ax_304 = fig.add_subplot(428, projection=map_304)
map_304.plot(axes=ax_304)
for contour in pos_contours:
    ax_304.plot_coord(contour, color="blue", linewidth=linewidth)
for contour in neg_contours:
    ax_304.plot_coord(contour, color="red", linewidth=linewidth)
```

What are we looking at?

What temperature/heights dominate these images?

Consider this plot. Approximately where would you place the dominate emission from

- HMI optical continuum
- AIA 1600 continuum
- AIA 304 (He II emission line)

?

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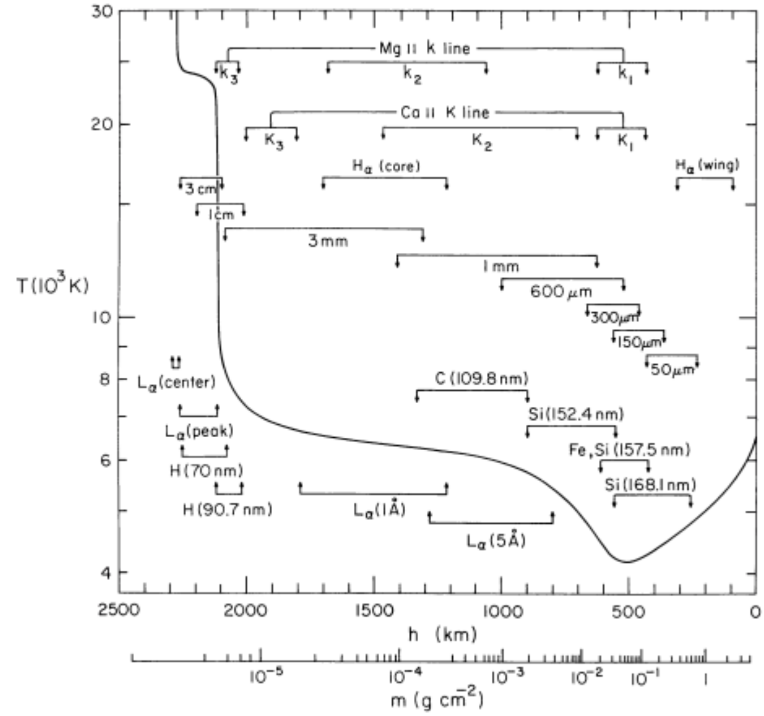

FIG. 1.—The average quiet-Sun temperature distribution derived from the EUV continuum, the Lα line, and other observations. The approximate depths where the various continua and lines originate are indicated.

In [ ]: